$$\lim_{\substack{x\to 0\\ y\to 0}} \frac{xy}{x^2+y^2}$$

$$\frac{a\ +\ b}{2} \quad \geq \sqrt{ab}$$

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

$$\begin{aligned} \nabla \cdot E &= \frac{\rho}{\varepsilon_0} \\ \nabla \cdot B &= 0 \\ \nabla \times E &= -\frac{\partial B}{\partial t} \\ \nabla \times B &= \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t} \end{aligned}$$

$$\begin{cases} \nabla \cdot E = \dfrac{\rho}{\varepsilon_0} \\ \nabla \cdot B = 0 \\ \nabla \times E = -\dfrac{\partial B}{\partial t} \\ \nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \dfrac{\partial E}{\partial t} \end{cases}$$

$$\oiint_{\Sigma} xdydz + ydzdx + zdxdy$$

$$\oiint_{\Sigma} xdydz + ydzdx + zdxdy$$

$$\begin{cases} y' + 3x^2 y &= x^2 \\ y\big|_{x=0} &= 1 \end{cases}$$